

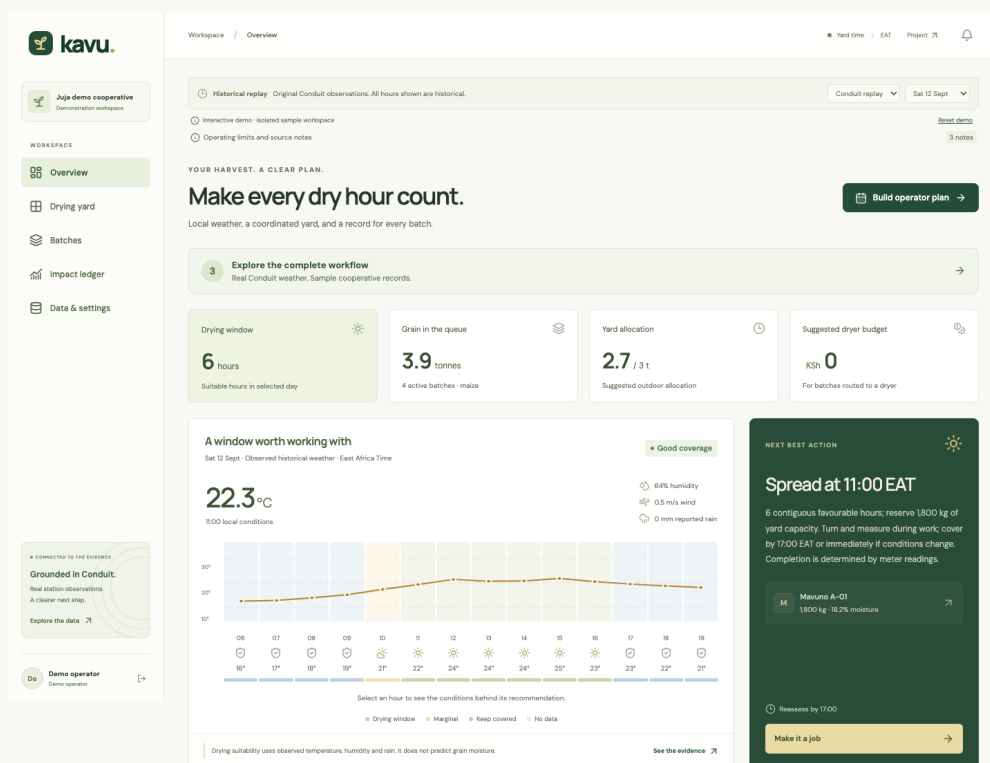
kavu.

Every dry hour counts.

A drying operations desk for maize cooperatives

A harvest takes months. At the drying yard, the next few hours matter.

Kavu helps a supervisor allocate limited drying space, record completed work and take the next moisture reading. Each lot keeps the decision and its result together.



Actual application screen. Historical Conduit replay with demonstration lots.

Shivam Gupta

Founder and product lead

kavu-drying.web.app

Data and the decision

18,364

unique observations

2,825

duplicate rows removed

The supplied exports span 28 August to 15 September 2026 UTC. A 144-hour gap remains missing. The interface shows East Africa Time. This is a retrospective replay for one JKUAT station, not a live warning or a nationwide observation network.

-
- 1** Original Conduit exports
GeoCSV files retain source attribution and checksums. The pipeline deduplicates timestamps and inspects available inputs.
 - 2** Evidence gates
Missing intervals remain missing. Humidity interpretation, rain counters and station health receive explicit handling. Operating thresholds need local validation.
 - 3** Shared decision logic
Environmental conditions, batch metadata and yard capacity produce an explained plan. An operator commits and completes tasks.
 - 4** Cloud accounts and a local option
Firebase Hosting serves React. Cloud Run runs the API with durable Firestore records. Express and SQLite remain available for local self-hosting.
 - 5** New measurement and record
Recent measured moisture determines eligibility for storage review. Task completion alone cannot certify grain quality or aflatoxin status.

Cloud release verified at kavu-drying.web.app. The historical plan uses observed weather retrospectively. It demonstrates the workflow and does not estimate forecast accuracy or drying rates.

A business worth testing

KSh 2,176

Drying-tariff equivalent of the example reduction

1.8 tonnes x 3.2 percentage points x KSh 377.80

Mavuno A-01 is an example 1,800 kg lot. A reading from 18.2% to 15.0% gives KSh 2,176.128, displayed rounded. The comparison uses NCPB's published tonne tariff and the intake-to-current moisture gap.

A transparent reference comparison. Measuring actual savings requires labour, transport, handling and alternative drying costs.

Prospective buyer

A cooperative or aggregator that already dries maize and uses a moisture meter. One supervisor coordinates many farmers' lots. The existing alternative is a weather report plus a notebook.

Price hypothesis

KSh 2,500

per active month per site.
Test support effort, daily use and willingness to renew.

Illustrative operating assumptions

Revenue per site per active month	KSh 2,500
Infrastructure allocation at 20 sites	KSh 150
Support and operating allowance	KSh 350
Contribution before sales, onboarding and founder pay	KSh 2,000

Planning assumptions only. At two sites, the same KSh 3,000 infrastructure budget costs KSh 1,500 per site. Seasonality, support and onboarding can erase the apparent margin.

Pilot, limits and sources

A supervised pilot with up to three maize yards

01 Observe current work

Document handling, representative sampling, handovers and actual receipts before changing the process.

02 Run in shadow mode

Compare suggestions with experienced operators. Record disagreements and retain existing protective practices.

03 Test continued use

Measure adoption, rechecks, administrative time and actual costs. Ask whether a buyer will renew at an agreed price.

Proposed research, with partner recruitment still to come. Initial histories test daily usefulness. Expansion depends on relevant weather coverage, local validation and confirmed data rights.

Sources and traceability

1. Conduit observations and source files

Official hackathon folder. Attribute 3d-fewsnets.icdp.ucar.edu, instrument 61. Confirm redistribution and commercial rights with the owner.

2. KALRO Maize Training of Trainers Manual, July 2021

Module 10 covers post-harvest handling, drying and moisture measurement. It does not validate Kavuu's operating thresholds.

3. NCPB drying service

Published reference: KSh 377.80 per tonne per percentage point. Actual depot terms and total costs may differ.

4. Kavuu repository and methodology

Raw-source checksums, quality report, decision logic, tests and the fuller market memo accompany the build.

Shivam Gupta supplied product direction, commercial focus and quality constraints. OpenAI Codex assisted research, engineering, test development and documentation. No field outcomes or additional human contributions are implied.